جامعة عفت
EFFAT UNIVERSITY
research
Student Scientific Research Forum & Exhibition

The 4th Undergraduate Research Forum

Elevating Societies Through Transformations

AI-Driven Multi-Disease Health Risk Prediction System

Research Brief

- Uses three real-world healthcare datasets
- Combines lifestyle factors and clinical indicators
- Applies advanced machine learning models
- Produces clear predictions and recommendations for users

Introduction

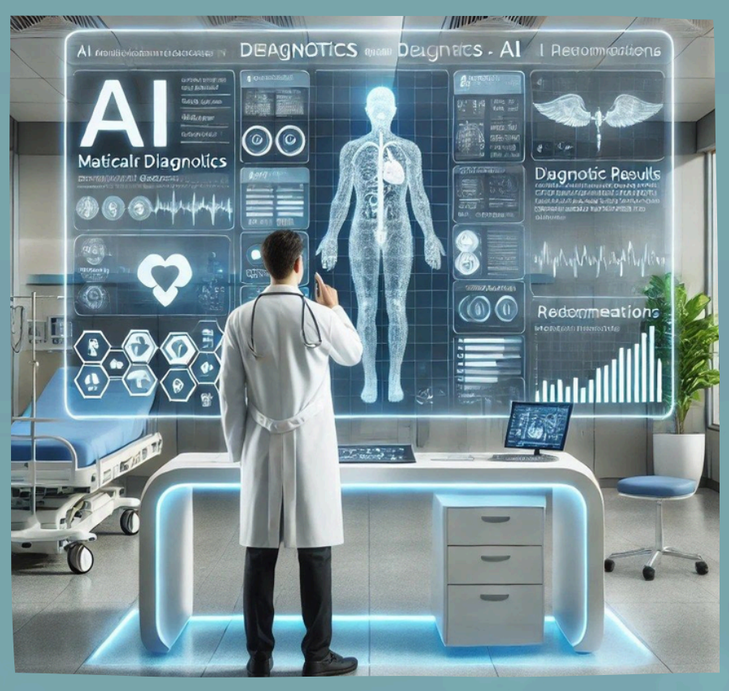
Chronic diseases such as diabetes, sleep disorders, and heart disease are increasing worldwide and often remain undetected until advanced stages.

This project leverages machine learning to analyze lifestyle and clinical data in order to predict health risks early, enabling preventive action and healthier decision-making.

Methodology and Design

1. **Data Understanding:**Reviewing datasets, features, and class distribution
2. **Data Cleaning & Preprocessing:** Handling missing values, encoding, scaling
3. **Exploratory Data Analysis (EDA):**Visualizing patterns and correlations
4. **Model Development:**Testing Logistic Regression, KNN, SVM, Random Forest, XGBoost
5. **Final Prediction System:**User input → Prediction → Risk probability → Recommendations







Significant Findings

- Combining lifestyle + clinical data improves prediction accuracy
- Random Forest and XGBoost achieved the best performance
- Handling data imbalance using SMOTE significantly enhanced model reliability
- Unified multi-disease prediction offers a holistic health assessment

Conclusion and Recommendations

This project demonstrates the effectiveness of AI in preventive healthcare by delivering a unified system for early health risk detection. While not a replacement for medical diagnosis, the system provides valuable insights that encourage:

- Healthier lifestyle choices
- Early medical consultation when needed
- ncreased health awareness

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